كلية الأمير سلطان بن عبدالعزيز
للخدمات الطبية الطارئة
Prince Sultan Bin Abdulaziz College for
Emergency Medical Services

جامعة الملك سعود
King Saud University
1957

Hand Hygiene

Public Health and Safety for EMS Specialists



Learning Objectives: What You Will Be Able to Do

1. Explain why hand hygiene is essential in infection prevention and EMS practice.
2. Differentiate between common hand hygiene agents.
3. Identify the advantages and limitations of alcohol-based hand rub and soap and water.
4. Explain when and how appropriate hand hygiene should be performed.
5. Differentiate between transient and resident skin flora.
6. Identify situations in EMS that require hand hygiene.
7. Explain why gloves do not replace hand hygiene.
8. Apply hand-hygiene principles to common paramedic scenarios.

Hand Hygiene Is the Most Effective Way to Prevent the Spread of Infection

- Recommended by the CDC (2002) and the WHO (2009) as the first line of infection prevention.
- This chapter follows Category IA and IB recommendations, the strongest level.
- Hand hygiene protects two people: the patient and the provider.
- It costs seconds. Missing it can cost a life.

WHAT DO IA AND IB MEAN?

IA: strongly recommended, supported by strong studies.

IB: strongly recommended, supported by some studies and strong reasoning.

Both are equally binding. The letter shows the strength of the evidence, not the strength of the rule.

40%

average hand hygiene compliance of healthcare workers in observational studies (range 5% to 81%)

ACTIVITY · Quick Vote

How many times did you clean your hands during your last shift?

Fewer than 5 · 5 to 15 · More than 15

Hands Touch Everything, and Microorganisms Travel With Them

Hands touch the patient, the equipment, and you
Every contact can pick up or leave microorganisms.

Microorganisms survive on hands for several minutes
Long enough to reach the next surface or the next patient.

Moisture increases transfer
Wet hands and wet skin areas move more microorganisms than dry ones.

More contacts per hour = more risk
Busy shifts create more opportunities and lower compliance.



EMS example: a patient with a chest infection coughs inside the ambulance for 20 minutes. Droplets land on the rails, the monitor, the oxygen tubing and your hands, and every surface you touch afterwards becomes part of the transmission route.

1847: Hand Washing Reduced Deaths Before Germs Were Discovered

GENERAL HOSPITAL OF VIENNA, AUSTRIA

DOCTORS' WARD

Doctors came directly from the autopsy room to deliver babies.

High death rate of mothers after childbirth.

MIDWIVES' WARD

Midwives did not perform autopsies.

Much lower death r

The intervention

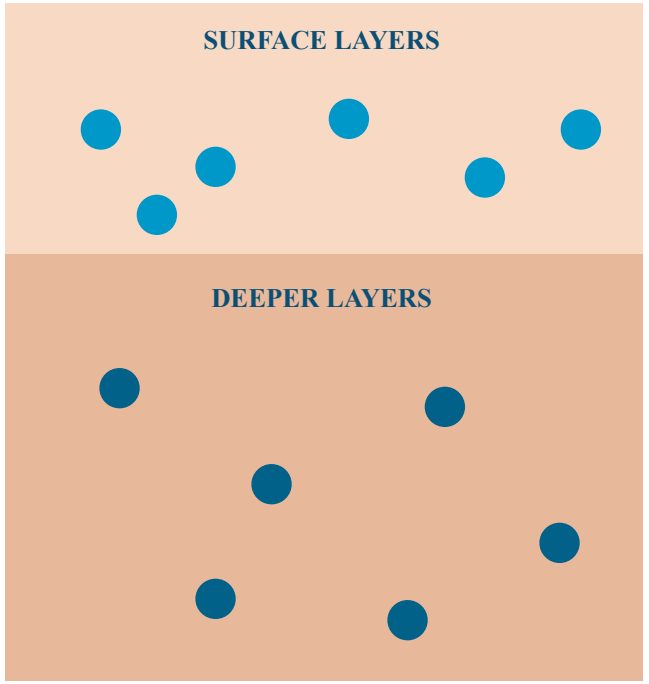
Han' tiseptic
s ry
hs

The evidence has been clear for more than 170 years. The difficulty is behaviour, not knowled

ACTIVITY · Think-Pair-Share

Why do you think the hospital rejected these results at the time? What does that tell us about changing behaviour today?

Transient Flora Is Easy to Remove and the Most Dangerous



	TRANSIENT FLORA	RESIDENT FLORA
Where	Surface layers of the skin	Attached to deeper layers
Removed by hand hygiene?	Yes, easily removed by hand washing	Difficult to remove
Link to healthcare-associated infection	Most often the cause	Less often the cause
In EMS terms	Picked up from the last patient, the stretcher or the monitor	Your own permanent skin bacteria; normal and protective

WHAT INCREASES TRANSFER OF TRANSIENT FLORA?

The type of microorganism · the number present · the level of moisture

Highest-risk body areas for colonisation: perineal and groin area, then armpits, trunk and upper limbs.

Contaminated Hands Can Carry Microorganisms Between Patients

FIVE STEPS ARE NEEDED FOR TRANSMISSION THROUGH HANDS

1. Microorganisms on the patient's skin, on surfaces, or on equipment near the patient
2. They transfer to the provider's hands
3. They survive on the hands for at least several minutes
4. Hand hygiene is missed, done badly, or done with the wrong agent
5. Contaminated hands touch the next patient, or equipment the next patient will touch

Step 4 is the only step the provider fully controls. Break it, and the chain stops.

ACTIVITY · Think-Pair-Share

A paramedic touches the stretcher rail after unloading a patient, adjusts the monitor, then meets the next patient. Which step of the chain would you break, and how?

Hand Hygiene Is One Term With Several Methods Inside It

Hand washing

Plain (non-antimicrobial) soap and water.

Removes dirt and transient flora; does not kill.

Antiseptic hand wash

Soap containing an antiseptic agent, with water.

Removes and kills.

Antiseptic hand rub

Alcohol-based product rubbed on all surfaces of the hands until dry. No water.

Fast, kills, no sink needed.

Surgical hand antisepsis

Antiseptic wash or rub of hands and forearms before sterile procedures.

Removes transient flora and reduces resident flora.

Visibly soiled hands

Hands with dirt you can see, or contamination with blood or body fluids (for example faeces or urine). Rule: visibly soiled hands need soap and water. An alcohol rub is not enough.

BEFORE WE COMPARE AGENTS

Hand-hygiene agents differ in effectiveness, safety, residual activity and skin tolerance.

Plain Soap Removes Microorganisms but Does Not Kill Them

WHAT IT DOES

- Works by detergent (cleaning) action, not by killing
- Removes dirt, soil and organic material such as blood
- Can remove transient flora from the hands
- The correct choice when hands are visibly soiled

LIMITATIONS

- Little or no antimicrobial activity
- May fail to remove pathogens
- Repeated washing can paradoxically increase bacterial counts on the skin
- Skin irritation and dryness with frequent use

EMS example

A trauma patient bleeds onto your gloves and wrist. After removing gloves, use soap and water first. The soap will not kill what is there, but it removes the blood, and blood must be removed before any antiseptic can work properly.

Alcohol-Based Hand Rub Works Quickly Against Many Microorganisms

HOW IT WORKS

- Ethanol or isopropanol denatures (breaks down) the proteins of microorganisms
- **Effective concentration: 60% to 95% alcohol mixed with water**
- **Above 95% is LESS effective because the reaction needs water**
- The most effective agent for routine hand hygiene by healthcare personnel

WHAT IT KILLS (LABORATORY DATA)

- Gram-positive and gram-negative bacteria, including multidrug-resistant organisms
- Mycobacterium tuberculosis
- Many fungi
- Most enveloped viruses: influenza, HIV, herpes simplex
- Hepatitis B and C: enveloped, but the evidence at 60–70% alcohol is less certain

ACTIVITY · Quick Vote

Which is stronger for hand hygiene: 70% alcohol or 99% alcohol?

Vote first. Then explain: without water, alcohol evaporates before it can denature proteins.

Alcohol Has Limits: Know When It Is Not Enough

ADVANTAGES

- Fast: works in seconds
- More effective than other agents for routine hand hygiene
- Best option against multidrug-resistant organisms
- Covers gram-positive, gram-negative, fungi and TB
- No sink needed, so it suits the field

LIMITATIONS

- **Poor against bacterial spores (Clostridioides difficile), protozoan cysts and non-enveloped viruses (for example norovirus)**
- No persistent or residual activity: the effect ends when the hands are dry
- Less effective when blood or organic material is present
- Flammable and volatile, which is a storage issue in a hot ambulance

Three situations where alcohol fails: visibly soiled hands · C. difficile · non-enveloped viruses such as norovirus. In all three, use soap and water.

Knowledge Check 1: Which Method?

SCENARIO

You transport an elderly patient from a nursing home with chronic diarrhoea and a documented history of Clostridioides difficile infection. At the hospital you remove your gloves. Your hands look clean. What is the best action?

A Apply alcohol-based hand rub for 20 seconds

B Wash hands with soap and water

C Apply alcohol-based hand rub twice

D No hand hygiene is needed because gloves were worn

Answer and rationale in the speaker notes. Ask students to vote before revealing.

Other Hand Hygiene Agents (1 of 2)

Agent	Advantages	Disadvantages	FDA position and cautions
Chlorhexidine gluconate (CHG)	Works well against gram-positive bacteria; effective against enveloped viruses; substantial residual activity; minimally affected by organic material such as blood	Less effective against gram-negative bacteria, fungi and non-enveloped viruses; minimally effective against M. tuberculosis; does not kill spores; inactivated by natural soaps and hand creams (anionic emulsifying agents)	Conjunctivitis and severe corneal damage reported with 1% chlorhexidine; keep away from the eyes
Chloroxylenol (PCMX)	Good activity against gram-negative bacteria, mycobacteria and certain viruses; only minimally affected by organic matter such as blood	Contradictory data on effectiveness; needs a longer time to work; less residual activity than chlorhexidine; neutralised by non-ionic surfactants	FDA: not enough information available to classify it as safe and effective
Hexachlorophene	Bacteriostatic; good activity against S. aureus; cumulative effect with repeated use; residual activity	Less effective against gram-negative bacteria, fungi and mycobacteria; only modestly effective after a single use; toxic, absorbed through the skin	FDA: not generally recognised as safe and effective as an antiseptic hand wash, especially for infant bathing

HOW TO COMPARE ANY AGENT

Ask four questions: what is its spectrum, how fast does it act, does it keep working after application (residual activity), and does blood or organic material stop it?

Other Hand Hygiene Agents (2 of 2)

Agent	Advantages	Disadvantages	FDA position and notes
Iodine and iodophors	Effective against gram-positive and gram-negative bacteria, some spore-forming bacteria, mycobacteria, viruses and fungi; most hand-hygiene formulations contain 7.5% to 10% povidone-iodine	Iodine causes skin irritation and discolouration; iodophor effectiveness depends on pH, temperature, exposure time and the amount and type of inorganic material present	Widely used; iodophors are less irritating than free iodine
Quaternary ammonium compounds	More effective against gram-negative bacteria; some activity against lipophilic viruses	Less effective against gram-positive bacteria; minimally effective against mycobacteria and fungi; effectiveness decreased by organic matter such as blood; not compatible with anionic detergents	FDA: several compounds have insufficient data to be classified as safe and effective for an antiseptic hand wash
Triclosan	Non-ionic, colourless, 0.2% to 2% in antimicrobial soaps; most effective against gram-positive organisms including MRSA; reasonably effective against mycobacteria and Candida; some persistent activity on the skin; not substantially affected by organic matter	Less effective against gram-negative organisms, especially Pseudomonas aeruginosa; affected by pH, surfactants, emollients, humectants and the ionic properties of the formulation	FDA 1994: insufficient data to classify as safe and effective; FDA inquiry into safety and efficacy, focused on products marketed to the general public (later status in notes)

ADDED FOR EMS 313

For EMS practice: chlorhexidine is the agent to remember, for its residual activity. Povidone-iodine is commonly used for skin preparation before injections and needs contact time; wiping and injecting immediately wastes it.

Alcohol-Based Hand Rub vs Soap and Water: Not Interchangeable

	ALCOHOL-BASED HAND RUB	SOAP AND WATER
Speed	20 to 30 seconds; no sink needed	40 to 60 seconds; needs sink and towels
Kills or removes?	Kills most bacteria, fungi, enveloped viruses	Removes mechanically; plain soap does not kill
Spores and non-enveloped viruses	**Poor (C. difficile, norovirus)**	**Removed by washing and rinsing**
Visibly soiled hands	**Not effective; blood reduces activity**	**Required**
Residual activity	None	None (plain soap)
Skin	Less irritating than frequent washing	Dryness and irritation with repeated use
In EMS	Between routine patient contacts; at the point of care; pocket bottle	After blood or body-fluid contact; diarrhoea calls; when hands are visibly dirty

Default in the field: alcohol-based hand rub. Switch to soap and water when hands are visibly soiled, or after C. difficile or norovirus exposure.

Hand Rub and Handwash Share the Same Six Rubbing Steps



WHO: How to handrub / How to handwash

ALCOHOL-BASED HAND RUB · 20 TO 30 SECONDS

1a, 1b Apply a palmful of the product in a cupped hand and cover all surfaces.

2 to 7 The six rubbing steps, then keep rubbing until the hands are dry.

SOAP AND WATER · 40 TO 60 SECONDS

0, 1 Wet hands with water, then apply enough soap to cover all surfaces.

2 to 7 The same six rubbing steps.

8, 9, 10 Rinse with water, dry thoroughly with a single-use towel, and use the towel to turn off the tap.

STEPS 2 TO 7, IDENTICAL IN BOTH

2 Palm to palm
3 Right palm over the back of the left hand, fingers interlaced, then the other side
4 Palm to palm with fingers interlaced
5 Backs of the fingers against the opposite palm, fingers interlocked
6 Rotational rubbing of each thumb held in the other palm
7 Rotational rubbing of the fingertips in the opposite palm, both hands

Technique Rules: Where Hand Hygiene Goes Wrong



MOST OFTEN MISSED

- Fingertips and under the nails (step 7)
- Thumbs (step 6)
- Between the fingers (steps 3 and 4)
- Backs of the hands and wrists

HAND RUB: TWO RULES

If your hands are dry in 10 seconds, you used too little.

Never wipe or shake the hands to speed up drying. The product must finish working.



HANDWASH: WHY STEPS 9 AND 10 MATTER

Wet hands transfer far more microorganisms than dry hands, so dry thoroughly with a single-use towel.

You touched the tap with contaminated hands before washing. Use the towel to turn it off, or the contamination comes back.

THE NUMBERS YOU WILL SEE

Hand rub: 20 to 30 seconds, until dry (WHO).

Handwash: 40 to 60 seconds for the whole procedure (WHO).

At least 15 seconds of rubbing is the CDC minimum quoted in the source deck.

Spot the Mistake: How Many Errors Can You Find?

ACTIVITY · Spot the Mistake

Ahmed finishes a trauma call. There is blood on his gloves.

He removes the gloves and wipes his hands with an alcohol wipe.

He dries his hands on his uniform trousers.

He picks up his personal phone and writes the patient report.

At the next call he puts on new gloves and begins the assessment.

After the shift he refills the half-empty hand-rub bottle in the ambulance drawer.



Instructions

Work in pairs. Three minutes.

Find every hand-hygiene error and say what Ahmed should have done instead.

Be ready to explain which link of the transmission chain each error creates.

What Is Surgical Hand Antisepsis?

WHAT IT IS

Pre-operative cleansing of the hands and forearms with antiseptic agents.

Removes transient flora and reduces resident flora.

MOST EFFECTIVE PRODUCTS

- Formulations of 60% to 95% alcohol alone
- 50% to 95% alcohol combined with a quaternary ammonium compound, hexachlorophene or chlorhexidine gluconate
- Waterless, alcohol-based rubs

WHY ADD CHLORHEXIDINE?

Alcohol kills fast but stops working when dry.

A long operation inside a warm, moist glove lets resident flora regrow. Chlorhexidine keeps working for hours.

Before putting on sterile gloves, use an antimicrobial soap or an alcohol-based hand rub with persistent activity. [CDC and WHO, IB]

ACTIVITY · Think-Pair-Share

Why are the forearms included in surgical hand antisepsis but not in routine hand hygiene?

Surgical Hand Antisepsis: Key Recommendations

WHO IB

Artificial nails are prohibited in settings that require surgical hand antisepsis

WHO IB

Brushes are not recommended

CDC and WHO IB

Scrub hands and forearms for the length of time suggested by the facturer

CDC and WHO IB

Long scrub times are not necessary

CDC IB

Before an alcohol-based surgical scrub with persistent activity, pre-

WHO IB

Use enough alcohol-based hand rub to keep hands and forearms

CDC and WHO IB

After applying the alcohol-based product, dry hands and forearms thoroughly before putting on sterile gloves

CDC and WHO IB

Use an antimicrobial soap or an alcohol-based hand rub with persistent activity before putting on sterile gloves

Why no brushes and no long scrubs? Scrubbing damages the skin and brings resident flora up from the deeper layers, so the bacterial count rises instead of falling. The product's contact time does the work.

Choose an Agent That Staff Will Use

IB

Low potential for irritation

Even when used many times per shift.

IB

Ask the staff

Feel, fragrance and skin tolerance decide whether people use it.

CDC IB · WHO II

Cost is not the main factor

A cheap product that dries the skin lowers compliance and costs more.

IA

Never top up a partly full dispenser

Contamination risk. Use pre-filled bags or bottles.

IA

Provide hand lotion or cream

Skin protection is part of infection control, but the cream must be compatible with the agent.

IB

Teach skin care

Education programmes should include how to reduce irritation and damage.

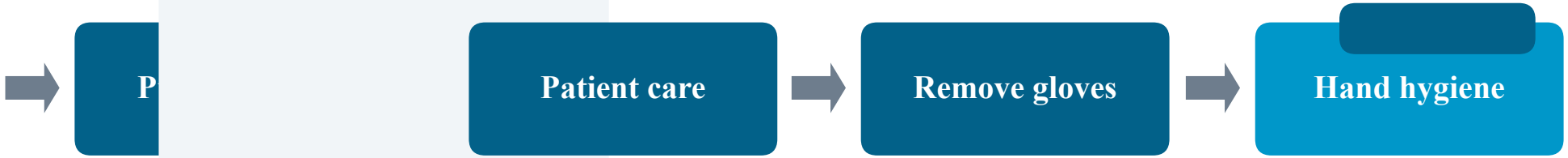
PRODUCT COMPATIBILITY MATTERS

Ask the manufacturer about the risk of product contamination [WHO IB] and about using hand lotions, creams or alcohol-based antiseptics together [CDC IB, WHO II].

EMS: a pocket-sized bottle per provider; bottles are never refilled in the ambulance drawer; a compatible hand cream is stocked at the station in dry, hot weather.

Gloves Do Not Replace Hand Hygiene

THE CORRECT SEQUENCE, EVER



WHY GLOVES ARE NOT ENOUGH

- Gloves can have tiny holes you cannot see
- Hands become contaminated while removing gloves
- The warm, moist space inside a glove lets bacteria multiply on your skin
- **WHO Category IB: the use of gloves does not reduce the need for hand hygiene**

FOUR GLOVE RULES (CDC AND WHO, IB)

- Remove gloves after caring for the patient
- Dispose of gloves after use
- **Never use the same pair for more than one patient**
- **Never wash gloves for re-use**
- Also: no artificial nails or extenders when in direct contact with patients (IB)

Knowledge Check 2: Gloves

SCENARIO

A paramedic transports two patients one after the other in the same ambulance. Between them she removes her gloves and immediately puts on a new pair, without performing hand hygiene. Which statement is correct?

A Acceptable, because the new gloves are clean, so the hands do not matter

B Acceptable, because the first patient had no known infection

C Not acceptable, because hand hygiene is required after removing gloves and before putting on new ones

D Acceptable, because the first pair was worn for less than ten minutes

Answer and rationale in the speaker notes. Ask students to vote before revealing.

The Contaminated Ambulance: Equipment Carries Microorganisms

INDIRECT TRANSMISSION

Patient → **Paramedic hand** → **Monit**



HIGH

Radio and tablet

Door handles and grab rails

Your personal phone

THE PHONE PROBLEM

A phone touched during patient care is rarely cleaned and goes home with you. Keep it out of the patient compartment, or disinfect it after every patient.

ACTIVITY · Mini Case Discussion

Vomiting patient; stretcher wiped, monitor cables not. Trace the route to the next patient (notes).

Why Hand Hygiene Compliance Fails

average 40%

SYSTEM

Sinks far away or too few · lack of soap and paper towels · lack of time · understaffed or overcrowded wards · agents that irritate and dry the skin

BELIEFS

'Gloves are enough' · patient seen as low risk · lack of knowledge of guidelines · forgetfulness · scepticism about the value · disagreement with the policy

CULTURE

No role model among peers or supervisors · no feedback · no institutional priority · no active promotion · no safety climate

WHAT OBSERVATION STUDIES SHOW

- Compliance is lower before patient contact than after. Providers protect themselves more than the patient.
- Compliance falls as the number of hand-hygiene opportunities per hour rises, which is when it matters most.
- Compliance is lower at night than by day, lower among physicians than nurses, and lower after a clean task than a dirty one.

What Improves Hand Hygiene Compliance

CDC and WH

Monitor and give feedback

Facilities should monitor compliance with hygiene practices and policies and feed the back to staff.

CDC and WHO IB

ty

stituti iority with nd fu

WHO IA

Education that targets the barriers

Programmes should address the factors found to influence compliance, not only the products.

WHO IA

Multimodal, with senior executives

Programmes should be multi-faceted and multimodal and include support from senior executives.

CDC IA

Multidisciplinary programmes

Implement multidisciplinary improvement programmes that include easy access to alcoho based hand rub.

Product where care happens

Hand rub at the room entrance, at the bedside, in convenient locations and in pocket-sized containers; dispensers at the point of care.

In EMS the point of care is the side of the stretcher: a pocket bottle for every provider and a dispenser mounted in the patient compartment, not in a cabinet in the cab.

When Should a Paramedic Perform Hand Hygiene?

BEFORE

- Patient contact
- IV insertion
- Medication preparation
- Airway procedures
- Handling clean equipment

AFTER

- Patient contact
- Blood or body-fluid exposure
- Removing gloves
- Touching contaminated equipment
- Cleaning the ambulance
- Contact with the patient environment

THE WHO 'FIVE MOMENTS', TRANSLATED TO EMS

1. Before touching the patient
2. Before a clean or aseptic procedure
3. After body-fluid exposure risk
4. After touching the patient
5. After touching the patient's surroundings

Breaking the Chain of Transmission Through Hand Hygiene

1. Microorganisms on the patient, surfaces or equipment
2. Transfer to the provider's hands
3. Survive on the hands for minutes
4. skin. You can always control step 4.

ies or how long microorganisms survive

Right method

Rub by default; wash when soiled, after C. difficile or norovirus

Right timing

Before as much as after; five moments, every call

Right technique

The six rubbing steps; until dry for the rub, 40 to 60 seconds at the sink

Common EMS Hand-Hygiene Mistakes and How to Fix Them

Changing gloves without ha

Fix: Hand hygiene before and after every pair.

lcohol rub on visibly bloody hands

Fix: Soap and water first; alcohol does not work through blood.

Touching a personal phone during patient care

Fix: Phone stays out of the patient compartment, or is disinfected after every patient.

Rubbing while walking, dry in 10 seconds

Fix: Enough product, all surfaces, until dry: about 30 seconds.

Wiping gloves to keep them on for the next task

Fix: Gloves are single use: remove, discard, hand hygiene.

Refilling half-empty bottles; storing rub in a hot cabin

Fix: Pre-filled bottles only; store away from heat.

ADDED FOR EMS 313 · NOT IN ORIGINAL CHAPTER 2

Practical EMS Scenarios: What Would You Do?

ACTIVITY · Group Scenario



Respiratory patient

A patient with pneumonia coughs continuously during a 20-minute transport. You adjust his oxygen mask twice and hold the rail on every turn.

Which moments of hand hygiene apply? Which surfaces need cleaning before the next patient?



IV insertion in a moving ambulance

You need an IV in a stable patient during transport. There is no sink. Your hands are not visibly soiled.

Which method, when, and what do you do with the gloves before and after?



Needlestick during a trauma call

While securing the IV in a bleeding patient, the needle pricks your finger through the glove.

What are your immediate actions in the first minute? What follows?

Three groups, one scenario each, five minutes. Present in two minutes: the actions, in order, and the reason for each.

Knowledge Check 3: The Patient Environment

SCENARIO

After a call, you unload the stretcher, wipe the rails, and adjust the monitor cables. You did not touch the patient at any point during the transport; your partner did. Before the next call, is hand hygiene required?

A No, because there was no direct patient contact

B No, because the rails were wiped with disinfectant

C Yes, because contact with the patient's surroundings and equipment requires hand hygiene

D Only if the patient had a known infection

Answer and rationale in the speaker notes. Ask students to vote before revealing.

Final Knowledge Check

4. A surgical team needs skin protection that keeps working through a three-hour operation inside sterile gloves. Which property of the hand-antisepsis product matters most?

- **A** Fast onset of action
- **B** Residual (persistent) activity, for example from chlorhexidine
- **C** A pleasant fragrance
- **D** The highest possible alcohol concentration

5. A colleague buys 99% alcohol because he believes it is stronger than the 70% hand rub supplied by the unit. What should you tell him?

- **A** He is right; a higher concentration always kills more
- **B** Concentrations above 95% are less effective because alcohol needs water to denature proteins
- **C** Both work the same; the difference is only price
- **D** 99% is better only for visibly soiled hands

Answers and rationale in the speaker notes.

كلية الأمير سلطان بن عبدالعزيز للخدمات الطبية الطارئة
Prince Sultan Bin Abdulaziz College for Emergency Medical Services
جامعة الملك سعود
King Saud University

Key Takeaways

- Hand hygiene is one of the most important infection-prevention practices.
- Hand hygiene protects the patient and the provider.
- Alcohol-based hand rub and soap and water are not interchangeable in every situation.
- Gloves do not replace hand hygiene.
- Hands and equipment can transmit microorganisms between patients.
- Hand hygiene is a clinical patient-safety intervention, not just a routine task.

"Clean hands protect the patient in front of you—and the patient you have not met yet."